Muhammad Umer Khan Niazi

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Summary

Second-year AI student at UET Lahore with experience building end-to-end NLP pipelines and deployable ML tools. Focused on practical machine learning, systems programming, and writing production-quality code.

Education

University of Engineering and Technology — Lahore, Pakistan
BS Artificial Intelligence | GPA: 3.25/4.0 — *2025 – Present*

Forman Christian College — Lahore, Pakistan
Intermediate Computer Science — *2022 – 2024*

Projects

News NLP Pipeline | *Python, BERTopic, scikit-learn, spaCy, Transformers, Streamlit* — 2026 – Present

- Processed 350k+ news articles and headlines from Dawn (2010–2025) through an end-to-end NLP pipeline
- Applied BERTopic with UMAP and KMeans for unsupervised topic discovery across 15 years of coverage
- Built downstream tasks including sentiment analysis, named entity recognition, and news category classification
- Deployed interactive Streamlit dashboard for exploring results, topic distributions, and visualizations

GB Wildlife Classifier | *Python, PyTorch, Streamlit* — 2026

- Trained a ResNet18 transfer learning pipeline to classify images of three endangered Himalayan species: brown bear, markhor, and snow leopard, achieving 98.16% validation accuracy
- Froze early layers and trained a custom 3-class fully connected head with augmentation to prevent overfitting on a limited dataset
- Stress-tested against out-of-distribution data and documented failure modes where visual and taxonomic similarity causes confident mispredictions
- Added an 80% confidence threshold to surface unknown inputs rather than force a classification

Shelter Outcome Predictor | *Python, scikit-learn, Pandas, Streamlit* — 2026

- Trained a Random Forest classifier on 62k+ shelter intake records to predict adoption, transfer, or euthanasia risk
- Applied balanced class weighting to prioritize recall on at-risk animals, achieving 63% euthanasia recall
- Deployed interactive Streamlit app for per-animal risk prediction with probability breakdown

WMUS | *Python, curses* — 2026 – Present

- Designed modular architecture separating audio playback, UI rendering, and state management
- Built a terminal music player for Windows with fuzzy search, vim-style navigation, and configurable keybindings
- Packaged with a versioned installer and fast metadata caching for large music libraries

Technical Skills

Languages: Python, C++, HTML, CSS
Tools & Platforms: Git, Linux, Streamlit, SQLite
Machine Learning & NLP: NumPy, Pandas, Matplotlib, scikit-learn, Seaborn, PyTorch, Transformers, Hugging Face, BERTopic